

Cem Ozcelik

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PROFESSIONAL SUMMARY

Senior Data Scientist with an Industrial Engineering background and nearly 5 years of progressive experience in end-to-end machine learning -- from data extraction and statistical modelling through containerized production deployment. Deep technical stack across Python, advanced SQL, PySpark/Databricks, XGBoost, and Docker-based MLOps. Industrial Engineering roots give a native cost-function lens (FP vs FN as business-cost asymmetry) -- directly applicable to credit risk, fraud, and anomaly detection. Sample impact: end-to-end Credit Risk pipeline (XGBoost + temporal holdout + Django + Docker on Lending Club); predictive maintenance & anomaly detection delivered a 15% increase in critical machine availability and 15% OEE improvement; ML-driven supply chain forecasting sustained an 8-hour weekly reduction in operational overtime; demand forecasting at 12.83% RMSPE on 1,115 stores. Comfortable communicating across business, BI, and engineering teams in Turkish and English. Demo: github.com/ccemozclk/Credit-Risk | github.com/ccemozclk/Rossmann-Demand-Forecasting

CORE COMPETENCIES

Skills: End-to-End ML Implementation, Statistical Modelling, Credit Risk & Anomaly Detection, Demand Forecasting, Feature Engineering, Class Imbalance & Temporal Validation, A/B Testing, Cross-Functional Communication (TR/EN)

Tools: Python (Advanced), SQL (Advanced), PySpark, Databricks, XGBoost, LightGBM, scikit-learn, TensorFlow, PyTorch, FastAPI, Django, Docker, Git, REST APIs, AWS (EC2/S3)

WORK EXPERIENCE

Anadolu Pet Akvaryum & Effeffe

Jan 2026 -- Present

Senior Database Analyst | Maltepe, Istanbul

- Architected and deployed **end-to-end ML pipelines** via versioned REST APIs on a Django + Docker platform -- ingestion, feature engineering, model training, and production serving all owned in-house.
- Directed predictive supply chain analytics using Python and advanced SQL, sustaining an **8-hour weekly reduction in operational overtime** through ML-driven demand forecasting.
- Translated SKU-level **Association Rule Mining** outputs into cross-sell strategy, driving continuous revenue growth -- bridging statistical modelling with business decision-making.
- Led cross-functional initiatives across BI, engineering, and operations, communicating model results clearly to non-technical stakeholders in Turkish and English.
- Tech:** Python, SQL, Django, REST APIs, Docker, PostgreSQL, ML pipelines, Association Rule Mining

Anadolu Pet Akvaryum & Effeffe

Sept 2024 -- Dec 2025

IT & Data Analytics Specialist | Maltepe, Istanbul

- Engineered predictive supply chain models in Python and PL/SQL, **reducing stockouts and improving inventory turnover** -- statistical forecasting applied to operational risk.
- Architected a full-stack Django web application from scratch (SOLID principles), establishing the foundation that later scaled into the ML deployment platform.
- Conducted **Route-Level Profitability Analysis** with advanced SQL, delivering a **3% improvement in overall logistics efficiency**.
- Tech:** Python, Django, PL/SQL, Oracle, SOLID, full-stack deployment

Birinci Automotive

Jan 2022 -- Sept 2024

Mid-Level Data Analyst | Cayirova, Kocaeli

- Developed a **Predictive Maintenance & Anomaly Detection framework** using unsupervised clustering on machine-sensor time-series -- the same statistical primitives that drive fraud detection and transaction anomaly modelling.
- Delivered a **15% weekly increase in critical machine availability** and a **15% improvement in Overall Equipment Effectiveness (OEE)** through cost-function-aware alerting (downtime cost >> false-alarm cost).
- Engineered robust **ETL pipelines in MSSQL and Python** for high-volume sensor data, replacing manual reporting with reliable analytical infrastructure.
- Tech:** Python, MSSQL, ETL, unsupervised clustering, time-series analysis, anomaly detection

Birinci Automotive

June 2021 -- Jan 2022

Junior Data Analyst | Cayirova, Kocaeli

- Conducted **Exploratory Data Analysis (EDA)** and **applied statistical methods** to workforce datasets, uncovering hidden patterns to optimize HR policies.
- Designed automated data workflows and ingestion pipelines, **improving reporting efficiency by 15%**.
- Tech:** Python, SQL, statistical analysis, ETL

PROJECTS

End-to-End Credit Risk Prediction MLOps & Financial Risk

Full lifecycle credit risk pipeline on Lending Club data: ingestion -> feature engineering -> model training -> containerized production app. **63% F1-Score** with XGBoost optimized for heavy class imbalance and **temporal holdout validation** (no data leakage -- the trap random splits hide in credit data). Interactive **Django EDA platform** lets non-technical reviewers explore data and model decisions. Architecture pattern transfers directly to fraud detection, payment intelligence, and behavioural anomaly modelling.

Python, XGBoost, CatBoost, Django, Docker, Pandas, scikit-learn | github.com/ccemozclk/Credit-Risk

Retail Demand Forecasting End-to-End MLOps Pipeline

~**12.83% RMSPE** on 6-week daily sales forecasting across 1,115 Rossmann stores using time-based validation (no temporal leakage). Object-oriented training pipeline (DataIngestion -> DataTransformation -> ModelTrainer) replacing static notebook code. **FastAPI REST inference + Streamlit dashboard with dynamic confidence intervals** translating ML metrics into business terms. Resilient inference layer handling unseen categorical inputs in production. Surfaced actionable insight: B-type stores yield lower promotion ROI vs A/C, enabling marketing budget reallocation.

Python, LightGBM, XGBoost, Optuna, FastAPI, Streamlit, Docker | github.com/ccemozclk/Rossmann-Demand-Forecasting

Scalable Recommendation Engine PySpark & Customer Segmentation

End-to-end personalized recommendation system on **Apache Spark / Databricks** (Online Retail II). **RFM analysis + K-Means** for behavioural customer segmentation into 5 strategic tiers (Champions, At-Risk, etc.). **ALS collaborative filtering** with implicit feedback and cold-start handling. Distributed processing primitives that scale 1:1 to banking customer-360 and transaction-graph use cases.

PySpark, Apache Spark, Databricks, K-Means, ALS, Streamlit | github.com/ccemozclk/Spark-Recommendation-Engine

EDUCATION

B.Sc. Industrial Engineering — Kirikkale University

Sept 2016 -- Feb 2021

Quantitative degree (Mathematics, Engineering, Statistics core) with rigorous foundation in statistical analysis, operations research, and optimization. Designed MILP models with IBM ILOG CPLEX; applied AHP and TOPSIS to multi-criteria decision problems -- direct prerequisite for cost-function-aware modelling in credit risk and anomaly detection.

CERTIFICATIONS

Associate Data Scientist Career Track (incl. PySpark, Recommendation Engines) — [DataCamp](#)

Jan -- May 2024

Data Science & Machine Learning Bootcamp (6 months, 10+ projects) — [Miuul](#)

Mar -- Sept 2022

SKILLS

Languages: Python (Advanced), SQL (Advanced), Bash

ML & Statistical: XGBoost, LightGBM, scikit-learn, h2o.ai, Statistical Modelling, A/B Testing, Class Imbalance Handling, Temporal Validation, Feature Engineering, Customer Segmentation (RFM, K-Means), Recommendation Systems (ALS, Item2Vec)

Deep Learning: TensorFlow, Keras, PyTorch, CNNs, RNNs, Computer Vision, Transfer Learning, Grad-CAM (XAI)

Big Data: PySpark, Apache Spark, Databricks, Distributed Computing

MLOps & Deployment: Docker, FastAPI, Django, Flask, REST APIs, Git, Linux, End-to-End ML Deployment, AWS (EC2, S3)

Databases: PostgreSQL, MSSQL, Oracle, ETL Pipelines, Pandas, NumPy

Advanced: Cost-Function Optimization, MILP, AHP/TOPSIS, OOP, System Architecture **Languages:** Turkish (Native), English (Fluent)